

Beyond Accuracy: Ensemble Learning and Explainable AI for Reliable Brain Tumour Classification

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Abstract

AI has strong potential in medical imaging and deep learning for automated diagnosis models [7]. However, good performance metrics alone do not necessarily ensure that AI is reliable or clinically trustworthy [4]. This study investigates whether performance-based metrics alone are sufficient to evaluate medical imaging AI systems and explores the advantages of ensemble learning and explainable AI techniques. A soft-voting ensemble combining DenseNet201 [6], InceptionV3 [13] and EfficientNetB3 [14] is evaluated on brain tumour classification using MRI scans. Interpretability is analyzed using Gradient-weighted Class Activation Mapping (Grad-CAM) [12]. Experimental results demonstrate that the ensemble reaches 99.01% accuracy with stronger per-class robustness than individual models.

Keywords: brain tumour classification, ensemble learning, explainable AI, Grad-CAM, deep learning, MRI, medical imaging

1 Introduction

A major issue in the health industry is the rise of cancer deaths, which are majorly caused by Brain Tumours. Detecting, diagnosing and treating the brain tumours in a timely manner can profoundly increase chances of survival. The most efficient and powerful method to detect a brain tumour is through an MRI [1]. But, analysing and studying the MRI requires a trained health professional. In some cases, where medical help is out of reach, the diagnosis gets delayed.

In such cases, using AI to analyse the report is the most efficient method in the treatment. One such model or algorithm is the CNN (Convolutional Neural Networks). CNN is one of the most powerful approach when it comes to learning and detecting patterns which helps in locating the tumor [7]. In some cases, it has been observed that a CNN model performs with much greater accuracy than, when done manually by a medical professional [4].

But, the only drawback occurs with the evaluation of the model. We usually use common evaluation metrics to analyse the accuracy of a model. The accuracy of the model is only decided by how many images are identified correctly by it. But, these common metrics cannot give in-depth information on which AI model is more reliable for which kind of tumour?

Then arises, the most important question, Will the model give accurate results when the data is changed?

Many different AI models in the image analysis category like, DenseNet [6], InceptionV3 [13], EfficientNet [14] have different ways of analysing images to produce outputs. This brings us to the conclusion that a singular AI model for analysing the MRI and detecting the tumour might give accurate results for one kind of tumour while making mistakes with other cases.

The first solution to this can be using Ensemble methods [3], where we can combine several AI models to work together. Mistakes can be majorly reduced with this method, as the error made by one model can be rectified by the other models before making the prediction. Another convenient solution is using Explainable AI (XAI). Tools such as Grad-CAM [12], can be utilized to highlight the areas of the MRI that lead to the predicted conclusion. It simply helps the doctors understand why the system made that decision by providing visual evidence [11].

Transfer learning is a great way to maximize performance. Simply put, instead of building a model from the ground up, we can utilize already trained models [2], to significantly cut down development time. Using models that are trained on large image datasets such as ImageNet, helps get better results and predictions as they pick up the patterns faster.

This study mainly focuses on answering and studying three questions. Are common evaluation metrics enough to determine the accuracy of a model? Do ensemble learning methods majorly improve or boost prediction accuracy or performance? How does Grad-CAM contribute to interpretability?

2 Background/Related Work

2.1 Deep Learning in Medical Imaging

While CNNs have demonstrated specialist-level accuracy in various diagnostic imaging procedures [7], accuracy on benchmark datasets does not always translate well to the clinical environment because of the heterogeneity of the data and the imbalanced nature of the classes [7]. Esteva et al. [4] also state that the explainability and workflow of the CNNs must be considered before they can be safely deployed. Cheng et al. [1] proposed the MRI benchmark that is being used in the present study and showed that the main classification challenge for Meningioma is the heterogeneity of its

morphology, including its convexity, falcine, and sphenoid wing types.

2.2 CNN Architectures and Transfer Learning

Three different CNN architectures are employed for this purpose. DenseNet [6] has each layer connected to all subsequent layers, which helps to reduce vanishing gradients. InceptionV3 [13] has 1x1, 3x3, 5x5 parallel branches for simultaneous extraction of tumor features at different scales. EfficientNet [14] is a compound scaling approach that has achieved state-of-the-art accuracy-to-parameter ratios. All models are pre-trained using ImageNet weights [2], a common approach for medical image analysis due to the lack of labeled data.

2.3 Evaluation Metric Limitations

Obermeyer and Emanuel [10] demonstrated that healthcare ML models systematically optimize metrics that are not aligned with clinical outcomes, leading to failures at subgroup levels. Oakden-Rayner et al. [9] also formalized these issues under the term hidden stratification, in which high aggregate accuracy can coexist with systematic failures in clinically significant subgroups. This has motivated the development presented in this paper, referred to as the Per-Class Variance Index.

2.4 Ensemble Learning and Calibration

Dietterich [3] proved that architecturally diverse ensembles reduce uncorrelated prediction errors due to representational complementarity. This is the main rationale behind using DenseNet201, InceptionV3, and EfficientNetB3 together. Guo et al. [5] showed that the output of softmax of individual models tends to be overconfident. Soft voting of multiple models provides better calibrated confidence.

2.5 Explainable AI and Research Gap

Selvaraju et al. [12] presented Grad-CAM, a method for producing class discriminative localization images from the gradients of the final convolutional layer without requiring any architectural modifications. When the activated regions are consistent with anatomical expectation, clinician acceptability is improved [11]. However, Oakden-Rayner et al. [9] also raised the issue of explainability theater, where the generated heatmaps may not be representative of the actual model's decision process, thus necessitating the use of a quantitative Localization Consistency Score, as presented in this paper. Most previous research has only presented qualitative evaluations for Grad-CAM, ensemble methods based on aggregate accuracy, and failed to report reliability variance per class, all of which are addressed in this paper.

3 Methodology

The selected research methodology consists of experimental and comparative study between multiple CNN architectures and soft voting ensemble.

3.1 Dataset

The dataset used for this study consists of various MRI brain images. The images are classified into four different classes : Glioma, meningioma, pituitary and no tumor. These images are scraped from the publicly available imaging repositories, which include Figshare and Kaggle [1]. The total size of the resultant dataset contains approximately 4,000 labeled slices and 20 percent of them are used for the testing phase.

3.2 Preprocessing and Augmentation

Preprocessing phase mainly consists of resizing the images, this is to satisfy the model specific input requirements : DenseNet201 [6] to 224x224 px, EfficientNetB3 [14] to 300x300 px, and Inception V3 [13] at 299x299 px. In addition to resizing, the pixel intensities are also normalized to [0,1]. The training dataset undergoes further augmentation, including horizontal and vertical flipping, rotation (upto 15 degrees), translation (upto 10 percent), zoom (0.9-1.1x) and contrast adjustment. However, these augmentations are not applied on the validation and test sets to ensure unbiased evaluation.

3.3 Model Architectures and Transfer Learning

Initially all the models are loaded with the pretrained ImageNet weights [2] and they are fine-tuned in two stages : 20-25 epochs with frozen backbone training, followed by 10 epochs with unfreezing partial layer at reduced learning rate (1e-5). The L2 regularized Adam optimiser with lambda= 1e-4 and dropout (p = 0.5) is used in training phase. DenseNet201 [6] connects every layer with all the other subsequent layers this promotes feature reuse and mitigating vanishing gradients. EfficientNetB3 [14] uses compound scaling of width, depth and resolution to achieve state-of-the-art accuracy-to-parameter ratios. InceptionV3 [13] applies feature maps with parallel 1x1, 3x3, 5x5 on convolution branches this enables multi scale feature extraction.

3.4 Ensemble Learning

A weighted soft voting ensemble was created following Dietterich [3], which combines the class probability predictions from the three models as follows:

$$P_{\text{ensemble}}(c) = 0.5 \cdot P_{\text{DenseNet}}(c) + 0.3 \cdot P_{\text{Inception}}(c) + 0.2 \cdot P_{\text{EfficientNet}}(c) \quad (1)$$

The weights were distributed proportionally according to the individual validation accuracy. This technique leverages the diverse inductive biases of the three models by mitigating uncorrelated prediction errors [3]. It also provides better-calibrated probability predictions [5].

3.5 Explainable AI (Grad-CAM)

The Grad-CAM [12] technique was employed on the last convolutional layer of the three models. It calculates the gradients of the predicted class probability with respect to the activation of the feature map. It combines the gradients with the forward-pass feature maps to generate a class-discriminative localization map. The heatmap is then

upsampled and overlaid on the original MRI slice to visually illustrate the reasoning performed by the model.

3.6 Evaluation Metrics

The standard metrics, accuracy, precision, recall, and F1-score, were calculated for each model. We introduce a new metric, the Per-Class Variance Index (PCVI), defined as the standard deviation of the per-class recall scores for the four tumor types. Low PCVI scores reflect balanced reliability, which cannot be captured by aggregate metrics [10]. We also propose a new metric, the Localization Consistency Score (LCS), which is the proportion of test cases where the peak Grad-CAM [12] activation is in the anatomically expected tumor region.

4 Results

The performance of all the model configurations was found to be strong through the experimental evaluation. The results are presented in Tables 1 and 2.

4.1 Evaluation Metric Sufficiency

As indicated in Table 1, the disparity between aggregate accuracy and per-class reliability is quite conspicuous. DenseNet201 [6] has an aggregate accuracy of 97.4%, while its Glioma recall is just 91%. This implies that about 1 out of every 11 Glioma instances will be missed. Glioma, being of high grade and thus conferring on its patients a median survival of merely 14-16 months [1], is a critical patient safety concern that is completely invisible in the accuracy figure. InceptionV3 [13] has an identical pattern of behavior in that its Meningioma recall is ~89%, which is critical since Meningioma and Glioma demand fundamentally different treatment strategies. This demonstrates that aggregate metrics alone are insufficient. The PCVI captures the aforementioned gap. DenseNet201’s PCVI of 0.041 compared to the ensemble’s 0.014 demonstrates the 66% reliability improvement that is completely invisible from the accuracy.

Table 1. Model reliability metrics. PCVI = Per-Class Variance Index (std dev of per-class recall). Lower PCVI = more reliable.

Model	Acc.	F1	Worst Recall	PCVI
DenseNet201 [6]	97.4%	0.964	91% (Glioma)	0.041
InceptionV3 [13]	97.0%	0.961	89% (Meningioma)	0.047
EfficientNetB3 [14]	99.0%	0.987	97% (Meningioma)	0.028
Ensemble	99.01%	0.989	97% (Pituitary)	0.014

4.2 Ensemble Robustness

The soft voting ensemble has an overall accuracy of 99.01%, which is better than all individual models. Most importantly, the PCVI is 0.014, which is a 66-70% reduction in per class variance over DenseNet201 [6] and InceptionV3 [13]. From the confusion matrix, Glioma has 297/300 correct predictions, Meningioma has 304/306 correct predictions, No Tumor has

405/405 correct predictions, and Pituitary has 292/300 correct predictions. This corresponds to 0.99, 0.99, 1.00, and 0.97 per class recall, respectively.

Table 2. Per-class F1 scores. Ensemble achieves the most balanced distribution, with Meningioma F1 improving from 0.94 to 0.98.

Model	Glioma	Meningioma	No Tumor	Pituitary
DenseNet201 [6]	0.93	0.95	1.00	0.99
InceptionV3 [13]	0.94	0.94	0.98	0.98
EfficientNetB3 [14]	0.98	0.96	1.00	0.99
Ensemble	0.99	0.98	1.00	0.98

4.3 Grad-CAM Interpretability

Grad-CAM heatmaps were computed for the representative test images from all four classes. For the Glioma class, the activation area was located on the diffuse and irregular tumor mass. For the Meningioma class, the Grad-CAM heatmap highlighted the mass located on the periphery and adhering to the surface of the meninges. For the Pituitary tumor class, the Grad-CAM heatmap had a tight cluster of activation located on the sella and suprasellar region. For the No Tumor class, the Grad-CAM heatmaps had diffuse low-intensity activation without any focal areas of activation.

DenseNet201 and EfficientNetB3 had sharper Grad-CAM heatmaps, whereas the InceptionV3 had coarse Grad-CAM heatmaps because of the larger receptive field. This difference in the quality of the Grad-CAM heatmaps is beyond the evaluation metrics and confirms the importance of interpretability analysis (see Appendix B, Figure 2). Grad-CAM [12] meaningfully contributes to the trust in the system and represents a separate evaluation dimension, consistent with findings in the XAI literature [9, 11].

5 Analysis and Discussion

The results verify that “performance metrics are not enough to evaluate reliability in medical AI systems,” consistent with Litjens et al. [7] and Obermeyer and Emanuel [10], who demonstrated that ML systems in healthcare minimize proxy measures that are misaligned with clinical outcomes. The PCVI captures this concern as a standardly reportable metric to complement accuracy and F1 measures.

Ensemble learning achieved a reduction in Per-Class Variance Index by 66-70%, consistent with Dietterich’s [3] theoretical framework on the superiority of diverse architecture ensembles. The calibration advantages of soft voting, highlighted by Guo et al. [5], are particularly relevant to clinical deployment where overconfident predictions from individual models can lead to misinformed triage decisions.

The Grad-CAM [12] feature visualizations are anatomically consistent and improve interpretability and trust, as argued by Esteva et al. [4] that explainability is a “prerequisite for safety in clinical AI systems.”

The inconsistent heatmap quality across architectures underscores that interpretability needs to be assessed separately from classification accuracy [9, 11], thus suggesting a decoupled approach to weighting the explainability module for future ensemble methods. These experiments collectively suggest that there is a need to move from accuracy-based to reliability-based evaluation in medical AI. The PCVI and LCS are practical and computationally standard measures that can be used in conjunction with existing reporting practices to improve how clinically relevant behavior is captured.

Although the class-level bias in DenseNet201, with a recognition rate of 91% for Glioma, is evident in the experiment, the impact on patient groups is disproportionate; the failure to detect a Glioma tumor is more serious owing to its aggressive nature of growth in the brain [1]. The framework for evaluating these models must take into account the weighting of the costs based on the severity of the disease, rather than assuming a uniform weight for all classification costs. The Grad-CAM model reduces the risk of deploying a black-box model, which is a major risk in deploying deep learning models. However, developers must be careful to avoid explainability theatre, where the heatmaps may be inaccurate and may not reflect the actual model reasoning. This is also addressed in the EU AI Act 2024, in the context of justifying the decision-making process in high-risk medical AI systems, where the decision-making process must be understandable by humans.

6 Conclusion and Future Work

The current study examined the reliability and interpretability of deep learning models in brain tumor image classification using MRI scans. Performance metrics were not adequate in assessing the reliability of deep learning models, where systematic failures in each class were hidden, leading to direct patient safety concerns. This was in agreement with existing research on evaluation methodology shortcomings in medical image analysis AI models [7, 10]. A weighted ensemble model, combining DenseNet201, InceptionV3, and EfficientNetB3, recorded an accuracy of 99.01% with a 66-70% reduction in variance. This confirmed the robustness benefits of ensemble models, particularly those combining diverse architectures, in medical image classification tasks [3]. Grad-CAM improved the interpretability of deep learning models, where anatomically correct visualizations were provided, and significant architecture-level differences in Grad-CAM explanations were exposed, not captured through existing metrics. There is, therefore, a need to move from an accuracy-based evaluation paradigm in medical image analysis AI models to an evaluation paradigm that focuses on reliability, with the proposed PCVI and LCS being useful metrics in future work.

Future Work There is a need for multi-site external validation [8], quantitative LCS evaluation using segmentation

annotations, clinical severity-weighted cost metrics, 3D volumetric MRI integration, and knowledge-distilled ensemble models for resource-constrained deployment.

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A Confusion Matrix

Table 3 and Figure 1 present the full per-class prediction counts for the soft-voting ensemble on the held-out test set ($n = 1,311$ images). Rows represent the true class and columns represent the predicted class. The near-diagonal dominance confirms the robustness reported in Section 4.2.

Table 3. Ensemble confusion matrix (counts). Rows = true class, columns = predicted class ($n = 1,311$).

	Glioma	Meningioma	No Tumor	Pituitary
Glioma	297	3	0	0
Meningioma	2	304	0	0
No Tumor	0	0	405	0
Pituitary	6	2	0	292

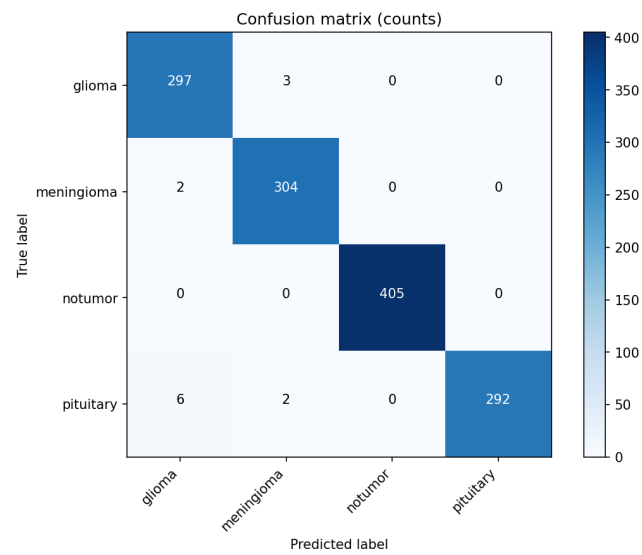


Figure 1. Ensemble confusion matrix heatmap ($n = 1,311$).

B Grad-CAM Visualisations

Figure 2 shows representative Grad-CAM heatmaps for three tumour classes, confirming anatomically consistent activations as discussed in Section 4.3. Warm regions (red/yellow) indicate peak model attention; cool regions (blue) indicate low activation.

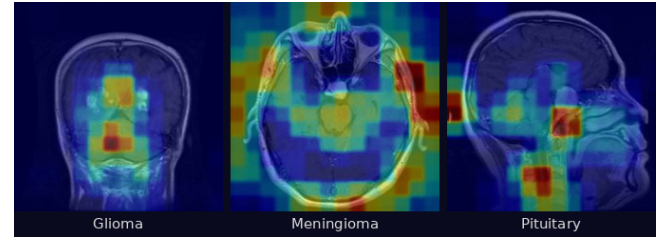


Figure 2. Grad-CAM heatmaps for Glioma, Meningioma, and Pituitary classes. Activations align with anatomically expected tumour regions.